Chinese_topic_Exp.3: Statistical Report

Fuyun WU

2019-06-17

Chinese topic processing

- Mandarin: 60id

Load relevant packages

```
library(plyr)
library(ggplot2)
library(reshape2)
library(scales)
library(MASS)
library(lme4)
```

```
## Loading required package: Matrix
```

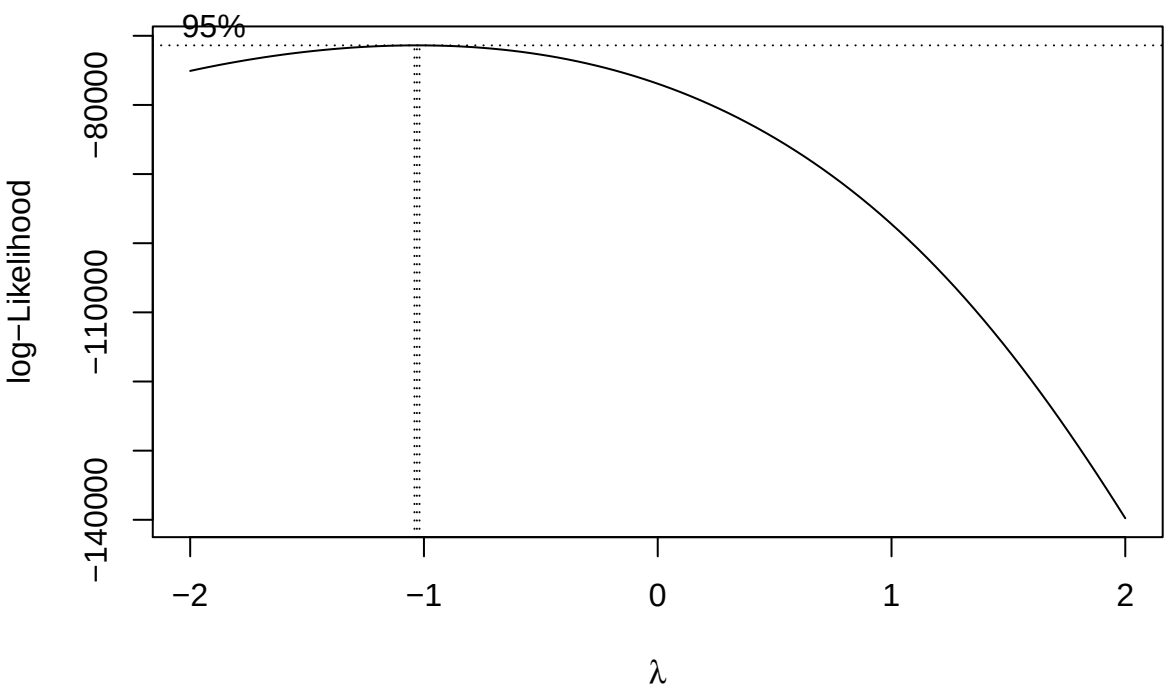
```
library(car)
library(languageR)
```

Import dataset: Exp3 topic w/ contexts

```
topic3 <- read.csv("topic3_w_context_60id.csv")
```

```
topic3$subj <- as.factor(topic3$subj)
topic3$item <- as.factor(topic3$item)
```

```
boxcox <- boxcox(topic3$RT~ topic3$condition * topic3$subj)
```



```r
##lamda peaks at -1, thus use rrt
topic3 <- subset(topic3, topic3$RT > 50 & topic3$ RT < 5000)
##eliminated 142 (0.82%) datapoints
topic3$rrt <- -1000/topic3$RT
```

sum-contrast coding

```r
order <- ifelse(topic3$condition%in%c("a","c"),-1,1)
complexity <- ifelse(topic3$condition%in%c("a","b"),-1,1)
orderxcomplexity <- ifelse(topic3$condition%in%c("b","c"),-1,1)

topic3$order <- order
topic3$complexity <- complexity
topic3$orderxcomplexity <- orderxcomplexity
head(topic3, n=5)
```

```
##   subj item condition order complexity constituent position Q   RT
## 1    1    1         a    -1         -1       comma       11 -  374
## 2    1    1         a    -1         -1         DCL       12 -  444
## 3    1    1         a    -1         -1          DP       13 - 1022
## 4    1    1         a    -1         -1       RC-V1       14 -  492
## 5    1    1         a    -1         -1       RC-V2       15 -  757
##          rrt orderxcomplexity
## 1 -2.6737968                1
## 2 -2.2522523                1
## 3 -0.9784736                1
## 4 -2.0325203                1
## 5 -1.3210040                1
```

RC-V2 (by part of speech):

```r
m0.RCV2 <- lmer(rrt ~ order + complexity + orderxcomplexity
                + (1+ order + complexity + orderxcomplexity||subj)
                + (1 + order + complexity + orderxcomplexity||item),
                subset(topic3,constituent =="RC-V2"))
summary(m0.RCV2)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic3, constituent == "RC-V2")
##
## REML criterion at convergence: 3832.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.8215 -0.6427 -0.0920  0.5754  3.8374
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.563217 0.75048
```

```
##  subj.1   order            0.010346 0.10171
##  subj.2   complexity       0.004823 0.06945
##  subj.3   orderxcomplexity 0.004764 0.06902
##  item     (Intercept)      0.012338 0.11107
##  item.1   order            0.000000 0.00000
##  item.2   complexity       0.000000 0.00000
##  item.3   orderxcomplexity 0.000000 0.00000
##  Residual                  0.705669 0.84004
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -3.06438    0.10194 -30.062
## order            -0.13805    0.02574  -5.364
## complexity        0.02217    0.02388   0.928
## orderxcomplexity  0.02598    0.02386   1.089
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```
m1.RCV2 <- lmer(rrt ~ order + complexity + orderxcomplexity
                + (1+ order + complexity + orderxcomplexity||subj)
                + (1 |item),
                subset(topic3,constituent =="RC-V2")) ##optimal model
summary(m1.RCV2)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + (1 | item)
##    Data: subset(topic3, constituent == "RC-V2")
##
## REML criterion at convergence: 3832.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.8215 -0.6427 -0.0920  0.5754  3.8374
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.563217 0.75048
##  subj.1   order            0.010346 0.10171
##  subj.2   complexity       0.004823 0.06945
##  subj.3   orderxcomplexity 0.004764 0.06902
##  item     (Intercept)      0.012337 0.11107
##  Residual                  0.705669 0.84004
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -3.06438    0.10194 -30.062
## order            -0.13805    0.02574  -5.364
```

```
## complexity        0.02217    0.02388   0.928
## orderxcomplexity  0.02598    0.02386   1.089
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

pos15 (RC-V2): W4 as reported in the paper

```r
m0.pos15.RCV2 <- lmer(rrt ~ order + complexity + orderxcomplexity
            + (1+ order + complexity + orderxcomplexity||subj)
             + (1 + order + complexity + orderxcomplexity||item),
         subset(topic3,position =="15"))
summary(m0.pos15.RCV2)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic3, position == "15")
##
## REML criterion at convergence: 3876.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.1551 -0.6031 -0.0611  0.5627  3.8624
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      4.968e-01 0.704817
##  subj.1   order            1.222e-02 0.110566
##  subj.2   complexity       5.569e-03 0.074624
##  subj.3   orderxcomplexity 7.506e-03 0.086635
##  item     (Intercept)      1.312e-02 0.114556
##  item.1   order            6.951e-03 0.083374
##  item.2   complexity       0.000e+00 0.000000
##  item.3   orderxcomplexity 4.269e-05 0.006534
##  Residual                  7.227e-01 0.850104
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -3.07883    0.09658  -31.88
## order            -0.15251    0.03155   -4.83
## complexity        0.01082    0.02439    0.44
## orderxcomplexity  0.01463    0.02507    0.58
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
```

```
## ordrxcmplxt 0.000  0.000 0.000
```

```r
m1.pos15.RCV2 <- lmer(rrt ~ order + complexity + orderxcomplexity
         + (1+ order + complexity + orderxcomplexity||subj)
       + (1 + order + orderxcomplexity||item),
      subset(topic3,position =="15")) ## optimal model
summary(m1.pos15.RCV2)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + orderxcomplexity |
##     item))
##    Data: subset(topic3, position == "15")
##
## REML criterion at convergence: 3876.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.1551 -0.6031 -0.0611  0.5627  3.8624
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      0.4967678 0.704818
##  subj.1   order            0.0122247 0.110565
##  subj.2   complexity       0.0055687 0.074623
##  subj.3   orderxcomplexity 0.0075057 0.086635
##  item     (Intercept)      0.0131232 0.114557
##  item.1   order            0.0069513 0.083374
##  item.2   orderxcomplexity 0.0000427 0.006534
##  Residual                  0.7226776 0.850104
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -3.07883    0.09658  -31.88
## order            -0.15251    0.03155   -4.83
## complexity        0.01082    0.02439    0.44
## orderxcomplexity  0.01463    0.02507    0.58
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

DE:

```r
m0.DE <- lmer(rrt ~ order + complexity + orderxcomplexity
       + (1+ order + complexity + orderxcomplexity||subj)
     + (1 + order + complexity + orderxcomplexity||item),
       subset(topic3,constituent =="DE"))
summary(m0.DE)
```

```
## Linear mixed model fit by REML ['lmerMod']
```

```
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic3, constituent == "DE")
##
## REML criterion at convergence: 3675.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -11.6708  -0.5729  -0.0411   0.5460   3.1351
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      4.394e-01 6.628e-01
##  subj.1   order            6.074e-03 7.793e-02
##  subj.2   complexity       0.000e+00 0.000e+00
##  subj.3   orderxcomplexity 0.000e+00 0.000e+00
##  item     (Intercept)      1.642e-02 1.281e-01
##  item.1   order            3.764e-03 6.135e-02
##  item.2   complexity       2.882e-03 5.368e-02
##  item.3   orderxcomplexity 6.568e-16 2.563e-08
##  Residual                  6.392e-01 7.995e-01
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -3.03091    0.09193  -32.97
## order            -0.18811    0.02649   -7.10
## complexity        0.04379    0.02375    1.84
## orderxcomplexity  0.03702    0.02107    1.76
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```r
m1.DE <- lmer(rrt ~ order + complexity + orderxcomplexity
        + (1+ order ||subj)
        + (1 + order + complexity + orderxcomplexity||item),
        subset(topic3,constituent =="DE"))
summary(m1.DE)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj)) + ((1 | item) + (0 + order | item) + (0 +
##     complexity | item) + (0 + orderxcomplexity | item))
##    Data: subset(topic3, constituent == "DE")
##
## REML criterion at convergence: 3675.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -11.6708  -0.5729  -0.0411   0.5460   3.1351
```

```
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.439362 0.66284
##  subj.1   order            0.006074 0.07793
##  item     (Intercept)      0.016417 0.12813
##  item.1   order            0.003764 0.06135
##  item.2   complexity       0.002882 0.05368
##  item.3   orderxcomplexity 0.000000 0.00000
##  Residual                  0.639220 0.79951
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -3.03091    0.09193  -32.97
## order            -0.18811    0.02649   -7.10
## complexity        0.04379    0.02375    1.84
## orderxcomplexity  0.03702    0.02107    1.76
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```
m2.DE <- lmer(rrt ~ order + complexity + orderxcomplexity
              + (1+ order ||subj)
              + (1 + order + complexity||item),
              subset(topic3,constituent =="DE")) ## optimal model
summary(m2.DE)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj)) + ((1 | item) + (0 + order | item) + (0 +
##     complexity | item))
##    Data: subset(topic3, constituent == "DE")
##
## REML criterion at convergence: 3675.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -11.6708  -0.5729  -0.0411   0.5460   3.1351
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  subj     (Intercept) 0.439362 0.66284
##  subj.1   order       0.006074 0.07793
##  item     (Intercept) 0.016417 0.12813
##  item.1   order       0.003764 0.06135
##  item.2   complexity  0.002882 0.05368
##  Residual             0.639220 0.79951
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
```

```
## (Intercept)      -3.03091    0.09193  -32.97
## order            -0.18811    0.02649   -7.10
## complexity        0.04379    0.02375    1.84
## orderxcomplexity  0.03702    0.02107    1.76
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

pos16,DE: W5 as reported in the paper

```
m0.pos16.DE <- lmer(rrt ~ order + complexity + orderxcomplexity
           + (1+ order + complexity + orderxcomplexity||subj)
        + (1 + order + complexity + orderxcomplexity||item),
        subset(topic3,position =="16"))
summary(m0.pos16.DE)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic3, position == "16")
##
## REML criterion at convergence: 3761.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -11.0799  -0.5498  -0.0229   0.5303   3.8153
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.399987 0.63245
##  subj.1   order            0.013670 0.11692
##  subj.2   complexity       0.005269 0.07259
##  subj.3   orderxcomplexity 0.000000 0.00000
##  item     (Intercept)      0.023349 0.15280
##  item.1   order            0.003817 0.06178
##  item.2   complexity       0.002214 0.04705
##  item.3   orderxcomplexity 0.000000 0.00000
##  Residual                  0.673526 0.82069
## Number of obs: 1438, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -3.03876    0.09004  -33.75
## order            -0.19596    0.02925   -6.70
## complexity        0.03594    0.02547    1.41
## orderxcomplexity  0.02917    0.02164    1.35
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
```

```
## order       0.000
## complexity  0.000  0.001
## ordrxcmplxt 0.000  0.001 0.001
```

```
m1.pos16.DE <- lmer(rrt ~ order + complexity + orderxcomplexity
      + (1+ order + complexity + orderxcomplexity||subj)
         + (1 + order + complexity ||item),
                    subset(topic3,position =="16"))
summary(m1.pos16.DE)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item))
##    Data: subset(topic3, position == "16")
##
## REML criterion at convergence: 3761.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -11.0799  -0.5498  -0.0229   0.5303   3.8153
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.399987 0.63245
##  subj.1   order            0.013670 0.11692
##  subj.2   complexity       0.005269 0.07259
##  subj.3   orderxcomplexity 0.000000 0.00000
##  item     (Intercept)      0.023349 0.15280
##  item.1   order            0.003817 0.06178
##  item.2   complexity       0.002214 0.04705
##  Residual                  0.673526 0.82069
## Number of obs: 1438, groups:  subj, 60; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -3.03876    0.09004  -33.75
## order            -0.19596    0.02925   -6.70
## complexity        0.03594    0.02547    1.41
## orderxcomplexity  0.02917    0.02164    1.35
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.001
## ordrxcmplxt 0.000  0.001 0.001
```

```
m2.pos16.DE <- lmer(rrt ~ order + complexity + orderxcomplexity
                    + (1+ order + complexity ||subj)
                    + (1 + order + complexity ||item),
                    subset(topic3,position =="16")) ## optimal model
summary(m2.pos16.DE)
```

```
## Linear mixed model fit by REML ['lmerMod']
```

```
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj)) + ((1 | item) +
##     (0 + order | item) + (0 + complexity | item))
##    Data: subset(topic3, position == "16")
##
## REML criterion at convergence: 3761.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -11.0799  -0.5498  -0.0229   0.5303   3.8153
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  subj     (Intercept) 0.399987 0.63245
##  subj.1   order       0.013670 0.11692
##  subj.2   complexity  0.005269 0.07259
##  item     (Intercept) 0.023349 0.15280
##  item.1   order       0.003817 0.06178
##  item.2   complexity  0.002214 0.04705
##  Residual             0.673526 0.82069
## Number of obs: 1438, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -3.03876    0.09004  -33.75
## order            -0.19596    0.02925   -6.70
## complexity        0.03594    0.02547    1.41
## orderxcomplexity  0.02917    0.02164    1.35
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.001
## ordrxcmplxt 0.000  0.001 0.001
```

HN:

```
m0.HN <- lmer(rrt ~ order + complexity + orderxcomplexity
        + (1+ order + complexity + orderxcomplexity||subj)
         + (1 + order + complexity + orderxcomplexity||item),
         subset(topic3,constituent =="HN"))
summary(m0.HN)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic3, constituent == "HN")
##
## REML criterion at convergence: 3530.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
```

```
## -4.2495 -0.6016 -0.0561  0.5770  4.0245
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      3.761e-01 6.133e-01
##  subj.1   order            5.328e-16 2.308e-08
##  subj.2   complexity       9.164e-03 9.573e-02
##  subj.3   orderxcomplexity 0.000e+00 0.000e+00
##  item     (Intercept)      1.127e-02 1.061e-01
##  item.1   order            8.462e-04 2.909e-02
##  item.2   complexity       6.777e-03 8.232e-02
##  item.3   orderxcomplexity 0.000e+00 0.000e+00
##  Residual                  5.767e-01 7.594e-01
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -3.12143    0.08449  -36.94
## order            -0.13405    0.02087   -6.42
## complexity        0.03487    0.02891    1.21
## orderxcomplexity  0.01472    0.02001    0.74
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```
m1.HN <- lmer(rrt ~ order + complexity + orderxcomplexity
              + (1+ order + complexity ||subj)
              + (1 + order + complexity ||item),
              subset(topic3,constituent =="HN"))
summary(m1.HN)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj)) + ((1 | item) +
##     (0 + order | item) + (0 + complexity | item))
##    Data: subset(topic3, constituent == "HN")
##
## REML criterion at convergence: 3530.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.2495 -0.6016 -0.0561  0.5770  4.0245
##
## Random effects:
##  Groups   Name        Variance  Std.Dev.
##  subj     (Intercept) 0.3761169 0.61328
##  subj.1   order       0.0000000 0.00000
##  subj.2   complexity  0.0091635 0.09573
##  item     (Intercept) 0.0112677 0.10615
##  item.1   order       0.0008463 0.02909
##  item.2   complexity  0.0067772 0.08232
##  Residual             0.5766675 0.75939
```

```
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -3.12143    0.08449  -36.94
## order            -0.13405    0.02087   -6.42
## complexity        0.03487    0.02891    1.21
## orderxcomplexity  0.01472    0.02001    0.74
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```
m2.HN <- lmer(rrt ~ order + complexity + orderxcomplexity
              + (1 + complexity ||subj)
              + (1 + order + complexity ||item),
              subset(topic3,constituent =="HN")) ## optimal model
summary(m2.HN)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     complexity | subj)) + ((1 | item) + (0 + order | item) +
##     (0 + complexity | item))
##    Data: subset(topic3, constituent == "HN")
##
## REML criterion at convergence: 3530.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.2495 -0.6016 -0.0561  0.5770  4.0245
##
## Random effects:
##  Groups   Name        Variance  Std.Dev.
##  subj     (Intercept) 0.3761170 0.61328
##  subj.1   complexity  0.0091635 0.09573
##  item     (Intercept) 0.0112677 0.10615
##  item.1   order       0.0008462 0.02909
##  item.2   complexity  0.0067773 0.08232
##  Residual             0.5766676 0.75939
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -3.12143    0.08449  -36.94
## order            -0.13405    0.02087   -6.42
## complexity        0.03487    0.02891    1.21
## orderxcomplexity  0.01472    0.02001    0.74
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

pos17.HN: W6 as reported in the paper

```
m0.pos17.HN <- lmer(rrt ~ order + complexity + orderxcomplexity
         + (1+ order + complexity + orderxcomplexity||subj)
       + (1 + order + complexity + orderxcomplexity||item),
         subset(topic3,position =="17"))
summary(m0.pos17.HN)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic3, position == "17")
##
## REML criterion at convergence: 3376.8
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.9760 -0.6470 -0.0594  0.5995  3.4994
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.401807 0.63388
##  subj.1   order            0.000000 0.00000
##  subj.2   complexity       0.007391 0.08597
##  subj.3   orderxcomplexity 0.005630 0.07503
##  item     (Intercept)      0.006134 0.07832
##  item.1   order            0.001488 0.03858
##  item.2   complexity       0.004155 0.06446
##  item.3   orderxcomplexity 0.000000 0.00000
##  Residual                  0.512683 0.71602
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -3.143860   0.085489  -36.77
## order            -0.156478   0.020446   -7.65
## complexity        0.012443   0.025541    0.49
## orderxcomplexity -0.007713   0.021210   -0.36
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```
m1.pos17.HN <- lmer(rrt ~ order + complexity + orderxcomplexity
          + (1+ complexity + orderxcomplexity||subj)
          + (1 + order + complexity ||item),
           subset(topic3, position =="17")) ## optimal model
summary(m1.pos17.HN)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
```

```
##     complexity | subj) + (0 + orderxcomplexity | subj)) + ((1 |
##     item) + (0 + order | item) + (0 + complexity | item))
##    Data: subset(topic3, position == "17")
##
## REML criterion at convergence: 3376.8
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.9760 -0.6470 -0.0594  0.5995  3.4994
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.401807 0.63388
##  subj.1   complexity       0.007391 0.08597
##  subj.2   orderxcomplexity 0.005630 0.07503
##  item     (Intercept)      0.006134 0.07832
##  item.1   order            0.001488 0.03858
##  item.2   complexity       0.004155 0.06446
##  Residual                  0.512683 0.71602
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -3.143860   0.085489  -36.77
## order            -0.156478   0.020446   -7.65
## complexity        0.012443   0.025541    0.49
## orderxcomplexity -0.007713   0.021210   -0.36
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

ADV:

```r
m0.ADV <- lmer(rrt ~ order + complexity + orderxcomplexity
       + (1+ order + complexity + orderxcomplexity||subj)
       + (1 + order + complexity + orderxcomplexity||item),
       subset(topic3,constituent =="ADV"))
summary(m0.ADV)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic3, constituent == "ADV")
##
## REML criterion at convergence: 3874.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -7.5554 -0.5863 -0.0344  0.5491  3.4764
```

```
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.440242 0.66351
##  subj.1   order            0.011566 0.10755
##  subj.2   complexity       0.017349 0.13172
##  subj.3   orderxcomplexity 0.007430 0.08619
##  item     (Intercept)      0.023231 0.15242
##  item.1   order            0.005083 0.07130
##  item.2   complexity       0.008924 0.09447
##  item.3   orderxcomplexity 0.000000 0.00000
##  Residual                  0.709756 0.84247
## Number of obs: 1438, groups:  subj, 60; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -3.054954   0.093803  -32.57
## order            -0.140670   0.029970   -4.69
## complexity        0.019373   0.033980    0.57
## orderxcomplexity  0.007505   0.024849    0.30
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.001
## ordrxcmplxt 0.000  0.001 0.001
```

```
m1.ADV <- lmer(rrt ~ order + complexity + orderxcomplexity
        + (1+ order + complexity + orderxcomplexity||subj)
        + (1 + order + complexity ||item),
         subset(topic3,constituent =="ADV")) ## optimal model
summary(m1.ADV)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item))
##    Data: subset(topic3, constituent == "ADV")
##
## REML criterion at convergence: 3874.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -7.5554 -0.5863 -0.0344  0.5491  3.4764
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.440242 0.66351
##  subj.1   order            0.011566 0.10755
##  subj.2   complexity       0.017349 0.13172
##  subj.3   orderxcomplexity 0.007430 0.08619
##  item     (Intercept)      0.023231 0.15242
##  item.1   order            0.005083 0.07130
##  item.2   complexity       0.008924 0.09447
```

```
##  Residual                  0.709756 0.84247
## Number of obs: 1438, groups:  subj, 60; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -3.054954   0.093803  -32.57
## order            -0.140670   0.029970   -4.69
## complexity        0.019373   0.033980    0.57
## orderxcomplexity  0.007505   0.024849    0.30
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.001
## ordrxcmplxt 0.000  0.001 0.001
```

pos18.ADV: W7 as reported in the paper

```
m0.pos18.ADV <- lmer(rrt ~ order + complexity + orderxcomplexity
        + (1+ order + complexity + orderxcomplexity||subj)
        + (1 + order + complexity + orderxcomplexity||item),
       subset(topic3,position =="18"))
summary(m0.pos18.ADV)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic3, position == "18")
##
## REML criterion at convergence: 3796.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.0542 -0.5885 -0.0385  0.5913  3.6595
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.450898 0.67149
##  subj.1   order            0.008916 0.09442
##  subj.2   complexity       0.012051 0.10978
##  subj.3   orderxcomplexity 0.009379 0.09685
##  item     (Intercept)      0.022882 0.15127
##  item.1   order            0.003491 0.05909
##  item.2   complexity       0.003702 0.06084
##  item.3   orderxcomplexity 0.000000 0.00000
##  Residual                  0.674045 0.82100
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -3.10574    0.09453  -32.85
## order            -0.19145    0.02761   -6.93
## complexity       -0.03141    0.02869   -1.09
```

```
## orderxcomplexity -0.04328    0.02499   -1.73
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```r
m1.pos18.ADV <- lmer(rrt ~ order + complexity + orderxcomplexity
           + (1+ order + complexity + orderxcomplexity||subj)
           + (1 + order + complexity ||item),
         subset(topic3,position =="18")) ## optimal model
summary(m1.pos18.ADV)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item))
##    Data: subset(topic3, position == "18")
##
## REML criterion at convergence: 3796.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.0542 -0.5885 -0.0385  0.5913  3.6595
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.450898 0.67149
##  subj.1   order            0.008916 0.09442
##  subj.2   complexity       0.012051 0.10978
##  subj.3   orderxcomplexity 0.009379 0.09685
##  item     (Intercept)      0.022882 0.15127
##  item.1   order            0.003491 0.05909
##  item.2   complexity       0.003702 0.06084
##  Residual                  0.674045 0.82100
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -3.10574    0.09453  -32.85
## order            -0.19145    0.02761   -6.93
## complexity       -0.03141    0.02869   -1.09
## orderxcomplexity -0.04328    0.02499   -1.73
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

quantifier:

```
m0.quantifier <- lmer(rrt ~ order + complexity + orderxcomplexity
        + (1+ order + complexity + orderxcomplexity||subj)
         + (1 + order + complexity + orderxcomplexity||item),
         subset(topic3,constituent =="quantifier"))
summary(m0.quantifier)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic3, constituent == "quantifier")
##
## REML criterion at convergence: 3414.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -12.3191  -0.5884  -0.0058   0.5898   3.2317
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      0.3353659 0.57911
##  subj.1   order            0.0000000 0.00000
##  subj.2   complexity       0.0040469 0.06362
##  subj.3   orderxcomplexity 0.0059141 0.07690
##  item     (Intercept)      0.0044582 0.06677
##  item.1   order            0.0029612 0.05442
##  item.2   complexity       0.0002561 0.01600
##  item.3   orderxcomplexity 0.0026874 0.05184
##  Residual                  0.5346135 0.73117
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -3.16026    0.07840  -40.31
## order            -0.09107    0.02224   -4.09
## complexity       -0.02290    0.02120   -1.08
## orderxcomplexity -0.02138    0.02412   -0.89
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```
m1.quantifier <- lmer(rrt ~ order + complexity + orderxcomplexity
        + (1+  complexity + orderxcomplexity||subj)
      + (1 + order + complexity + orderxcomplexity||item),
       subset(topic3,constituent =="quantifier")) ## optimal model
summary(m1.quantifier)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
```

```
##     complexity | subj) + (0 + orderxcomplexity | subj)) + ((1 |
##     item) + (0 + order | item) + (0 + complexity | item) + (0 +
##     orderxcomplexity | item))
##    Data: subset(topic3, constituent == "quantifier")
##
## REML criterion at convergence: 3414.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -12.3191  -0.5884  -0.0058   0.5898   3.2317
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      0.3353662 0.57911
##  subj.1   complexity       0.0040469 0.06362
##  subj.2   orderxcomplexity 0.0059141 0.07690
##  item     (Intercept)      0.0044582 0.06677
##  item.1   order            0.0029612 0.05442
##  item.2   complexity       0.0002561 0.01600
##  item.3   orderxcomplexity 0.0026874 0.05184
##  Residual                  0.5346134 0.73117
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -3.16026    0.07840  -40.31
## order            -0.09107    0.02224   -4.09
## complexity       -0.02290    0.02120   -1.08
## orderxcomplexity -0.02138    0.02412   -0.89
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

pos19.quantifier: W8 as reported in the paper

```
m0.pos19.quantifier <- lmer(rrt ~ order + complexity + orderxcomplexity
          + (1+ order + complexity + orderxcomplexity||subj)
          + (1 + order + complexity + orderxcomplexity||item),
          subset(topic3,position =="19"))
summary(m0.pos19.quantifier)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic3, position == "19")
##
## REML criterion at convergence: 3476.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -12.1238  -0.5421  -0.0081   0.5588   3.3788
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      0.3378235 0.58123
##  subj.1   order            0.0000000 0.00000
##  subj.2   complexity       0.0006315 0.02513
##  subj.3   orderxcomplexity 0.0000000 0.00000
##  item     (Intercept)      0.0062854 0.07928
##  item.1   order            0.0000000 0.00000
##  item.2   complexity       0.0005610 0.02369
##  item.3   orderxcomplexity 0.0000000 0.00000
##  Residual                  0.5717606 0.75615
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                   Estimate Std. Error t value
## (Intercept)      -3.131705   0.079305  -39.49
## order            -0.062517   0.019926   -3.14
## complexity        0.005662   0.020760    0.27
## orderxcomplexity  0.007180   0.019926    0.36
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```
m1.pos19.quantifier <- lmer(rrt ~ order + complexity + orderxcomplexity
             + (1+ complexity ||subj)
            + (1 + complexity ||item),
        subset(topic3,position =="19"))
summary(m1.pos19.quantifier)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     complexity | subj)) + ((1 | item) + (0 + complexity | item))
##    Data: subset(topic3, position == "19")
##
## REML criterion at convergence: 3476.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -12.1238  -0.5421  -0.0081   0.5588   3.3788
##
## Random effects:
##  Groups   Name        Variance  Std.Dev.
##  subj     (Intercept) 0.3378238 0.58123
##  subj.1   complexity  0.0006315 0.02513
##  item     (Intercept) 0.0062854 0.07928
##  item.1   complexity  0.0005610 0.02369
##  Residual             0.5717606 0.75615
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
```

```
##                   Estimate Std. Error t value
## (Intercept)      -3.131705   0.079305  -39.49
## order            -0.062517   0.019926   -3.14
## complexity        0.005662   0.020760    0.27
## orderxcomplexity  0.007180   0.019926    0.36
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

main-V1: maximal model overfitted; use Order as the fixed effect

```
xtabs(~order+condition,
      subset(topic3, constituent == "main-V1"))
```

```
##      condition
## order   a   b   c   d
##    -1   0   0 360   0
##    1    0 360   0   0
```

```
xtabs(~complexity+condition,
      subset(topic3, constituent == "main-V1"))
```

```
##           condition
## complexity   a   b   c   d
##         -1   0 360   0   0
##         1    0   0 360   0
```

```
m0.MV1 <- lmer(rrt ~ order + complexity + orderxcomplexity
    + (1+ order + complexity + orderxcomplexity||subj)
    + (1 + order + complexity + orderxcomplexity||item),
  subset(topic3,constituent =="main-V1"))
```

```
## fixed-effect model matrix is rank deficient so dropping 2 columns / coefficients
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model is nearly unider
##  - Rescale variables?
```

```
## fail to converge, and fixed-effect matrix shows that
## 2 coefficients complexity and orderxcomplexity were dropped.
## So it's best to use "order" as the only fixed effect

m1.MV1 <-lmer(rrt ~ order
              + (1+ order||subj)
              + (1 + order ||item),
              subset(topic3,constituent =="main-V1"))
summary(m1.MV1) # no effect of word order at MV1 (t=-0.134)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + ((1 | subj) + (0 + order | subj)) + ((1 | item) +
##     (0 + order | item))
##    Data: subset(topic3, constituent == "main-V1")
##
## REML criterion at convergence: 1657.5
##
```

```
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.5664 -0.5764 -0.0216  0.6303  2.8474
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  subj     (Intercept) 0.350509 0.59204
##  subj.1   order       0.018175 0.13481
##  item     (Intercept) 0.005133 0.07165
##  item.1   order       0.005697 0.07548
##  Residual             0.454132 0.67389
## Number of obs: 720, groups:  subj, 60; item, 24
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept) -2.952959   0.081771  -36.11
## order       -0.004584   0.034220   -0.13
##
## Correlation of Fixed Effects:
##       (Intr)
## order 0.000
```

check by positions: pos=20 (W9 as reported in the paper)

```r
m0.pos20.MV1 <- lmer(rrt ~ order + complexity + orderxcomplexity
          + (1+ order + complexity + orderxcomplexity||subj)
           + (1 + order + complexity + orderxcomplexity||item),
      subset(topic3,position =="20"))
summary(m0.pos20.MV1)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic3, position == "20")
##
## REML criterion at convergence: 3142.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.2029 -0.5733 -0.0013  0.6169  3.0619
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      0.3175097 0.56348
##  subj.1   order            0.0071484 0.08455
##  subj.2   complexity       0.0057961 0.07613
##  subj.3   orderxcomplexity 0.0019533 0.04420
##  item     (Intercept)      0.0026687 0.05166
##  item.1   order            0.0000000 0.00000
##  item.2   complexity       0.0044478 0.06669
##  item.3   orderxcomplexity 0.0002788 0.01670
##  Residual                  0.4352465 0.65973
```

```
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -2.89125    0.07553  -38.28
## order             0.03879    0.02053    1.89
## complexity        0.04337    0.02417    1.79
## orderxcomplexity  0.06171    0.01861    3.32
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```
m1.pos20.MV1 <- lmer(rrt ~ order + complexity + orderxcomplexity
           + (1+ order + complexity + orderxcomplexity||subj)
           + (1 + complexity + orderxcomplexity||item),
            subset(topic3,position =="20"))
summary(m1.pos20.MV1)  ### pos20: interaction=3.32, unpack it
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + complexity | item) + (0 + orderxcomplexity |
##     item))
##    Data: subset(topic3, position == "20")
##
## REML criterion at convergence: 3142.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.2029 -0.5733 -0.0013  0.6169  3.0619
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      0.3175094 0.56348
##  subj.1   order            0.0071484 0.08455
##  subj.2   complexity       0.0057961 0.07613
##  subj.3   orderxcomplexity 0.0019533 0.04420
##  item     (Intercept)      0.0026687 0.05166
##  item.1   complexity       0.0044478 0.06669
##  item.2   orderxcomplexity 0.0002788 0.01670
##  Residual                  0.4352465 0.65973
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -2.89125    0.07553  -38.28
## order             0.03879    0.02053    1.89
## complexity        0.04337    0.02417    1.79
## orderxcomplexity  0.06171    0.01861    3.32
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
```

```
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

main-V2: maximal model overfitted; use Order as the fixed effect

```
m0.MV2 <- lmer(rrt ~ order + complexity + orderxcomplexity
      + (1+ order + complexity + orderxcomplexity||subj)
     + (1 + order + complexity + orderxcomplexity||item),
    subset(topic3,constituent =="main-V2"))
```

```
## fixed-effect model matrix is rank deficient so dropping 2 columns / coefficients
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control
## $checkConv, : unable to evaluate scaled gradient
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control
## $checkConv, : Model failed to converge: degenerate Hessian with 1 negative
## eigenvalues
```

```
## fail to converge, and fixed-effect matrix shows that
## 2 coefficients complexity and orderxcomplexity were dropped.
## So it's best to use "order" as the only fixed effect

m1.MV2 <-lmer(rrt ~ order
              + (1+ order||subj)
              + (1 + order ||item),
              subset(topic3,constituent =="main-V2"))
summary(m1.MV2) #no effect of word order at MV2 (t=-1.05)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + ((1 | subj) + (0 + order | subj)) + ((1 | item) +
##     (0 + order | item))
##    Data: subset(topic3, constituent == "main-V2")
##
## REML criterion at convergence: 1656.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.8450 -0.5542  0.0042  0.6237  4.6559
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  subj     (Intercept) 0.499613 0.70683
##  subj.1   order       0.001101 0.03318
##  item     (Intercept) 0.007353 0.08575
##  item.1   order       0.018098 0.13453
##  Residual             0.446041 0.66786
## Number of obs: 720, groups:  subj, 60; item, 24
##
## Fixed effects:
##             Estimate Std. Error t value
## (Intercept) -1.86690    0.09619 -19.408
## order       -0.03933    0.03731  -1.054
##
## Correlation of Fixed Effects:
```

```
##       (Intr)
## order 0.000
```

check pos21 (W10 as reported in the paper)

```
m0.pos21.MV1 <- lmer(rrt ~ order + complexity + orderxcomplexity
        + (1+ order + complexity + orderxcomplexity||subj)
        + (1 + order + complexity + orderxcomplexity||item),
        subset(topic3,position =="21"))
summary(m0.pos21.MV1)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item) + (0 + orderxcomplexity | item))
##    Data: subset(topic3, position == "21")
##
## REML criterion at convergence: 3232.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.7769 -0.5515 -0.0046  0.5920  4.4726
##
## Random effects:
##  Groups   Name             Variance  Std.Dev.
##  subj     (Intercept)      4.080e-01 6.388e-01
##  subj.1   order            6.948e-18 2.636e-09
##  subj.2   complexity       5.604e-03 7.486e-02
##  subj.3   orderxcomplexity 1.167e-02 1.080e-01
##  item     (Intercept)      1.112e-02 1.054e-01
##  item.1   order            3.675e-03 6.062e-02
##  item.2   complexity       6.077e-03 7.795e-02
##  item.3   orderxcomplexity 0.000e+00 0.000e+00
##  Residual                  4.516e-01 6.720e-01
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -1.89794    0.08705 -21.803
## order            -0.09556    0.02160  -4.423
## complexity       -0.05623    0.02569  -2.188
## orderxcomplexity -0.03103    0.02254  -1.377
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

```
m1.pos21.MV1 <- lmer(rrt ~ order + complexity + orderxcomplexity
        + (1+ order + complexity + orderxcomplexity||subj)
        + (1 + order + complexity ||item),
       subset(topic3,position =="21"))
```

```r
summary(m1.pos21.MV1)  ### ME of order (t=-4.423), ME of complexity (t=-2.188)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: rrt ~ order + complexity + orderxcomplexity + ((1 | subj) + (0 +
##     order | subj) + (0 + complexity | subj) + (0 + orderxcomplexity |
##     subj)) + ((1 | item) + (0 + order | item) + (0 + complexity |
##     item))
##    Data: subset(topic3, position == "21")
##
## REML criterion at convergence: 3232.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.7769 -0.5515 -0.0046  0.5920  4.4726
##
## Random effects:
##  Groups   Name             Variance Std.Dev.
##  subj     (Intercept)      0.408036 0.63878
##  subj.1   order            0.000000 0.00000
##  subj.2   complexity       0.005604 0.07486
##  subj.3   orderxcomplexity 0.011667 0.10801
##  item     (Intercept)      0.011116 0.10543
##  item.1   order            0.003675 0.06062
##  item.2   complexity       0.006077 0.07795
##  Residual                  0.451599 0.67201
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##                  Estimate Std. Error t value
## (Intercept)      -1.89794    0.08705 -21.803
## order            -0.09556    0.02160  -4.423
## complexity       -0.05623    0.02569  -2.188
## orderxcomplexity -0.03103    0.02254  -1.377
##
## Correlation of Fixed Effects:
##             (Intr) order cmplxt
## order       0.000
## complexity  0.000  0.000
## ordrxcmplxt 0.000  0.000 0.000
```

Unpack the interaction detected at W9, using nested contrast

```r
topic3$TSV_inMC <- ifelse(topic3$condition=="d",-1,
                          ifelse(topic3$condition=="c",1,0))
topic3$TSV_inRC <- ifelse(topic3$condition=="b",-1,
                          ifelse(topic3$condition=="a",1,0))

topic3$MC_inSVO <- ifelse(topic3$condition == "b",-1,
                          ifelse(topic3$condition == "d",1,0))
topic3$MC_inTSV <-ifelse(topic3$condition =="a",-1,
                         ifelse(topic3$condition == "c",1,0))
head(topic3, n=5)
```

```
##   subj item condition order complexity constituent position Q   RT
```

```
## 1    1    1         a    -1         -1       comma       11 -  374
## 2    1    1         a    -1         -1         DCL       12 -  444
## 3    1    1         a    -1         -1          DP       13 - 1022
## 4    1    1         a    -1         -1       RC-V1       14 -  492
## 5    1    1         a    -1         -1       RC-V2       15 -  757
##          rrt orderxcomplexity TSV_inMC TSV_inRC MC_inSVO MC_inTSV
## 1 -2.6737968                1        0        1        0       -1
## 2 -2.2522523                1        0        1        0       -1
## 3 -0.9784736                1        0        1        0       -1
## 4 -2.0325203                1        0        1        0       -1
## 5 -1.3210040                1        0        1        0       -1
```

nested comparision pos20.MV1:

```
xtabs(~TSV_inMC + condition,subset(topic3,constituent =="main-V1"))
```

```
##         condition
## TSV_inMC   a   b   c   d
##        0   0 360   0   0
##        1   0   0 360   0
```

```
xtabs(~TSV_inRC + condition,subset(topic3,constituent =="main-V1"))
```

```
##         condition
## TSV_inRC   a   b   c   d
##       -1   0 360   0   0
##       0    0   0 360   0
```

```
xtabs(~order + condition, subset(topic3,constituent =="main-V1"))
```

```
##      condition
## order   a   b   c   d
##    -1   0   0 360   0
##    1    0 360   0   0
```

```
m0.pos20.MCV1.rrt.nested <-lmer(rrt ~ TSV_inMC + TSV_inRC + complexity
             + (1 + TSV_inMC + TSV_inRC + complexity||subj)
           + (1 + TSV_inMC + TSV_inRC + complexity||item),
        subset(topic3,position =="20"), REML = FALSE)

summary(m0.pos20.MCV1.rrt.nested)  ## rrt: t= -3.38, TSV slower than SVO in MC
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula:
## rrt ~ TSV_inMC + TSV_inRC + complexity + ((1 | subj) + (0 + TSV_inMC |
##     subj) + (0 + TSV_inRC | subj) + (0 + complexity | subj)) +
##     ((1 | item) + (0 + TSV_inMC | item) + (0 + TSV_inRC | item) +
##         (0 + complexity | item))
##    Data: subset(topic3, position == "20")
##
##      AIC      BIC   logLik deviance df.resid
##   3147.4   3215.9  -1560.7   3121.4     1427
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.1883 -0.5766 -0.0061  0.6145  3.1574
##
```

```
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  subj     (Intercept) 0.312237 0.55878
##  subj.1   TSV_inMC    0.011677 0.10806
##  subj.2   TSV_inRC    0.005081 0.07128
##  subj.3   complexity  0.005509 0.07422
##  item     (Intercept) 0.002580 0.05079
##  item.1   TSV_inMC    0.002135 0.04620
##  item.2   TSV_inRC    0.000000 0.00000
##  item.3   complexity  0.004354 0.06599
##  Residual             0.434418 0.65910
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##             Estimate Std. Error t value
## (Intercept) -2.89125    0.07492  -38.59
## TSV_inMC    -0.10050    0.02978   -3.37
## TSV_inRC     0.02293    0.02623    0.87
## complexity   0.04337    0.02398    1.81
##
## Correlation of Fixed Effects:
##            (Intr) TSV_MC TSV_RC
## TSV_inMC   0.000
## TSV_inRC   0.000  0.000
## complexity 0.000  0.000  0.000
```

```
xtabs(~MC_inSVO + condition, subset(topic3,constituent =="main-V1"))
```

```
##         condition
## MC_inSVO   a   b   c   d
##       -1   0 360   0   0
##       0    0   0 360   0
```

```
xtabs(~MC_inTSV + condition, subset(topic3,constituent =="main-V1"))
```

```
##         condition
## MC_inTSV   a   b   c   d
##        0   0 360   0   0
##        1   0   0 360   0
```

```
xtabs(~complexity + condition, subset(topic3,constituent =="main-V1"))
```

```
##           condition
## complexity   a   b   c   d
##         -1   0 360   0   0
##         1    0   0 360   0
```

```
m0.pos20.MCV1.rrt.nested.2<-lmer(rrt~ MC_inSVO + MC_inTSV + order
                + (1 + MC_inSVO + MC_inTSV + order||subj)
              + (1 + MC_inSVO + MC_inTSV + order||item),
            subset(topic3,position =="20"), REML = FALSE)

summary(m0.pos20.MCV1.rrt.nested.2) ## rrt: t=2.996, in SVO, MC faster than RC
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula:
## rrt ~ MC_inSVO + MC_inTSV + order + ((1 | subj) + (0 + MC_inSVO |
```

```
##     subj) + (0 + MC_inTSV | subj) + (0 + order | subj)) + ((1 |
##     item) + (0 + MC_inSVO | item) + (0 + MC_inTSV | item) + (0 +
##     order | item))
##    Data: subset(topic3, position == "20")
##
##      AIC      BIC   logLik deviance df.resid
##   3145.7   3214.2  -1559.8   3119.7     1427
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.1883 -0.5745 -0.0085  0.6270  3.0952
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  subj     (Intercept) 0.313053 0.55951
##  subj.1   MC_inSVO    0.012193 0.11042
##  subj.2   MC_inTSV    0.001997 0.04469
##  subj.3   order       0.006783 0.08236
##  item     (Intercept) 0.002945 0.05426
##  item.1   MC_inSVO    0.010175 0.10087
##  item.2   MC_inTSV    0.000000 0.00000
##  item.3   order       0.000000 0.00000
##  Residual             0.434386 0.65908
## Number of obs: 1440, groups:  subj, 60; item, 24
##
## Fixed effects:
##             Estimate Std. Error t value
## (Intercept) -2.89125    0.07511  -38.49
## MC_inSVO     0.10509    0.03508    3.00
## MC_inTSV    -0.01834    0.02523   -0.73
## order        0.03879    0.02036    1.90
##
## Correlation of Fixed Effects:
##          (Intr) MC_SVO MC_TSV
## MC_inSVO 0.000
## MC_inTSV 0.000  0.000
## order    0.000  0.000  0.000
```